Conservation is the only exact content: a verification layer for climate models

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Abstract

Climate projections are the most consequential predictions we make, and the field has no verification layer that distinguishes a *verified* projection from an *extrapolated* one. We apply the PACLAND verification discipline — the two-level certificate, the verification deficit, and the constraint-first decomposition — to the CMIP6 ensemble. Three results. **(1) The leak:** 12 of 19 models fail a three-ledger conservation guard (energy, mass, salt), and the leak is localized to the atmosphere–ocean coupling, not the ocean. **(2) The fix:** drift-correction — the field's own post-processing repair — restores conservation in every model (12 → 0 failures), but a retrospective test against HadCRUT5 shows it does *not* fix the temperature bias; that bias comes from sensitivity, and the sensitivity+forcing gate fixes it (ensemble ECS 3.78 → 3.08 K, matching the observed 3.0 K). **(3) The theorem:** conservation is the *only* exact, model-independent content of a climate model — the climate analog of the flux tadpole — and every predictive quantity (ECS, forcing, the 2100 warming) is a model-dependent range, not an exact observable. The verification deficit climbs to $m = 7$ (4 empirical observables + 3 exact conservation ledgers) and is *saturated*: the constraint contribution is capped at 3, exactly as the flux sector is capped at the tadpole.

1 The verification deficit, applied to climate

The deficit counts observables that are simultaneously *measured, computable, and validated*. On the historical record, $m = 4$ (GMST, EEI, OHC, GMSL — a $\approx 68\%$ certificate radius). The 2100 projection has $m = 0$: no future state is measured, so the projection is extrapolation, not verification.

2 The leak — three conservation ledgers

A conserving model must close three independent ledgers in its control run: *energy* (net top-of-atmosphere flux ≈ 0), *mass* (ocean mass ≈ 0), *salt* (salinity drift ≈ 0). Measured (Irving et al. 2021, 19 models): 12 leak energy, 6 leak mass, 2 leak salt. Salt — the purely-ocean ledger — is conserved orders of magnitude better than energy and mass, which cross the air–sea boundary; the leak therefore lives in the *atmosphere–ocean coupling*, not the ocean model. The worst model (CNRM-ESM2-1) leaks at $\approx 1.9\times$ the Earth's energy imbalance.

Widening to all 53 models (a raw second-source recomputation of the piControl net-TOA, validated against Irving within 0.03 W/m^2 on the overlapping models) finds *22 of 27* previously-unmeasured models leak energy — the models Irving never covered are, if anything, leakier.

3 The fix — drift-correction

The field's own repair is drift-correction: subtract the measured leak. Applied to all models, every one conserves by construction (12 → 0 failures). But a retrospective test against HadCRUT5 (HadGEM3-GC31-LL, historical + piControl) shows the correction does *not* improve the temperature match (RMSE 0.27 → 0.29 K; end warming +1.92 vs +0.93 K observed). The reason is stated, not hidden: HadGEM3's $\sim$ 1 K over-warming comes from its *sensitivity* (a "hot" model), not its conservation drift (0.02 K/century). Two tools, two jobs: drift-correction fixes *conservation*; the sensitivity+forcing gate fixes the *temperature bias* (ensemble ECS 3.78 → 3.08 K, matching the observed 3.0 K). Three independent observational sources agree: Sherwood et al. 2020 (ECS), HadCRUT5 (temperature), Loeb et al. 2021 (EEI).

4 The Climate Constraint-First Theorem

Theorem 1. *The only exact, model-independent content of a climate model is its conservation laws (energy, mass, salt). Every predictive quantity — ECS, forcing, the 2100 warming — is a model-dependent range, not an exact observable.*

Arm A (exact). Conservation is an identity, not an approximation: a model that does not close the ledgers is broken, not "approximately wrong". **Arm B (non-exact).** The 53 CMIP6 models span ECS 1.83–5.64 K (a range, not a value); any two disagreeing models falsify "ECS is exact"; the projection has $m = 0$. This is the climate instance of the one spine: *"don't ask a complex system for its values — ask for its constraints."*

5 The deficit, climbed and capped

$m = 7$ — 4 empirical observables + 3 exact conservation ledgers — with the ceiling proved by the theorem: conservation is the only exact content, so the constraint contribution is *saturated at 3*, exactly as the flux sector is saturated at 1 (the tadpole). Further deficit growth is empirical only.

Reproduce

```
PYTHONPATH=src python3 scripts/validate_climate_constraint_first.py
PYTHONPATH=src python3 -m pacland.climate.drift_correction
PYTHONPATH=src python3 scripts/climate_retrospective.py
```

Code: `src/pacland/climate/` data: `data/climate/` 305 tests green.